

THE MARGINAL PROPENSITY TO INVEST: EVIDENCE FROM ITALIAN PUBLIC DEBT SETTLEMENTS

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- ▶ **What is the Marginal Propensity to Invest (MPI) of firms?**
 - ◊ Response of firms' investment to an unexpected €1 increase in cash flow
- ▶ Important for theory & policy:
 - I MPI=0 in q -theory, but large with financial frictions $\rightarrow$ direct vs indirect channels?
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- ▶ **Yet, little causal evidence!** Challenge: isolate cash flow shocks from other shocks
(demand, productivity, ...)

THIS PAPER

I Natural experiment + merged administrative data:

- ◇ 2013 reform forcing Italian public administration to clear their debt to firms
- ◇ Universe of Italian public procurement contracts + firms' balance sheet & income statements

II Shift-share approach:

- ◇ Firm exposure to different public administrations receiving different resource allocation
- ◇ Cash flow ↑ for more exposed firms, demand unchanged

III Heterogeneous firm model:

- ◇ Calibrate financial frictions to interpret our estimates

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Marginal Propensity to Invest $\approx$ 50% annually

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Response stronger for **highly leveraged** & **young** firms & **capital-intensive** sectors

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Can replicate heterogeneity: **stabilization-efficiency** tradeoff ?

Marginal propensities & heterogeneity in macro

- ▶ **MPC as a central object:** (McKay-Reis, 2016; Kaplan-Moll-Violante, 2016; Bilbiie, 2020; Michelacci-Paciello-Pozzi, 2021; Auclert-Rognlie-Straub, 2024; Bilbiie-Galaasen-Gürkaynak-Maehlum-Molnar, 2025; Commault, 2025; Auclert-Monnery-Rognlie-Straub, 2026; Rachel-Ravn, 2026; Adams-Matthes, 2026; Broer-Kramer-Mitman, 2026)
- ▶ **Firm side comparatively underdeveloped:** (Ottonello-Winberry, 2020; Bilbiie-Melitz, 2025; Winberry-Auclert-Rognlie-Straub, 2025; Anagnostopoulos-Atesagaoglu-CarcelesPoveda-Giannitsarou, 2026; Bellifemine-Couturier-Tozzo, 2026)

We estimate the MPI, the MPC's analogue for firms

Investment-cash-flow sensitivity & financial frictions

- ▶ **Open debate:** (Fazzari-Hubbard-Petersen, 1988 vs Kaplan-Zingales, 1997; Erickson-Whited, 2000)
- ▶ **Challenge: cash flows correlate with investment prospects** (Commault & Martin-Baillon, 2026)

Natural experiment → shock to liquidity in broad cross-section of firms

PLAN FOR TODAY

1. Theoretical framework
2. Natural experiment
3. Empirical Analysis
4. Back to the model

THEORETICAL FRAMEWORK

A CANONICAL HETEROGENEOUS-FIRM MODEL

► Production: $y = zf(k, n)$ with DRS $\rightarrow$ profits $\pi(z, k) = \max_n zf(k, n) - wn$

$$v_t(z, k, b) = \max_{\iota, b', d} d + R^{-1} \mathbb{E}_t [(1 - \theta)v_{t+1}(z', k', b') + \theta v_{t+1}^{\text{exit}}]$$

$$\begin{aligned} \text{s.t. } d &= \pi(z, k) - \iota - \Phi(\iota, k) + R^{-1}b' - b + T_t \\ k' &= \iota + (1 - \delta)k \end{aligned}$$

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$$1 + \iota_0(z, k, b) = \mathbb{E}_0 \sum_{s=1}^{\infty} \frac{(1 - \delta)^{s-1}}{R^s} \frac{\partial \pi(z_s, k_s)}{\partial k}$$

- Frictionless q -theory $\rightarrow MPI(z, k, b) = \frac{\partial \iota_0}{\partial T_t} = 0$

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$$\text{s.t. } d = \pi(z, k) - \iota - \Phi(\iota, k) + R^{-1}b' - b + T_t$$

$$k' = \iota + (1 - \delta)k, \quad d \geq 0 \text{ } [\lambda], \quad b' \leq \chi k' \text{ } [\eta]$$

$$1 + \iota_0(z, k, b) = \mathbb{E}_0 \sum_{s=1}^{\infty} \frac{(1 - \delta)^{s-1}}{R^s} \left(\frac{\partial \pi(z_s, k_s)}{\partial k} \frac{\lambda(z_s, k_s, b_s)}{\lambda(z, k, b)} + \frac{\chi \eta(z_s, k_s, b_s)}{\lambda(z, k, b)} \right)$$

- Frictionless q -theory $\rightarrow MPI(z, k, b) = \frac{\partial \iota_0}{\partial T_t} = 0$

- Add financial frictions $\rightarrow MPI(z, k, b) = \frac{\partial \iota_0}{\partial T_t} > 0$

SEQUENCE SPACE JACOBIANS

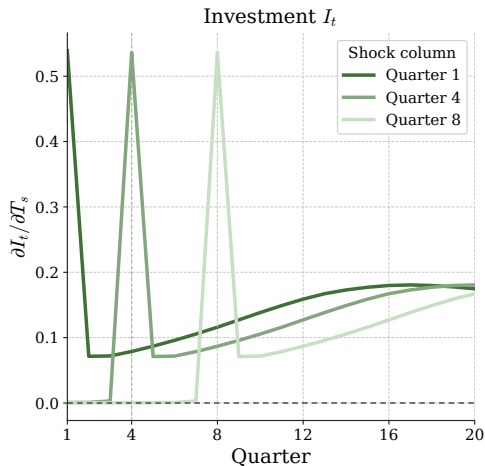
- ▶ Endogeneous distribution of firms $\mu_t(z, k, b)$

- ▶ Define aggregate investment function:

$$I_t = \int \iota(z, k, b; \{T_h\}_{h \geq t}) d\mu_t(z, k, b) = \mathcal{I}_t(\{T_h\}_{h \geq 0})$$

- ▶ Define the "iMPI" matrix

$$M_{t,s} = \frac{\partial \mathcal{I}_t(\{T_h\}_{h \geq 0})}{\partial T_s}$$



NATURAL EXPERIMENT

INSTITUTIONAL CONTEXT: THE 2013 SBLOCCA DEBITI

- ▶ Italy: chronic delays in public payments to suppliers (≈ 6 months)

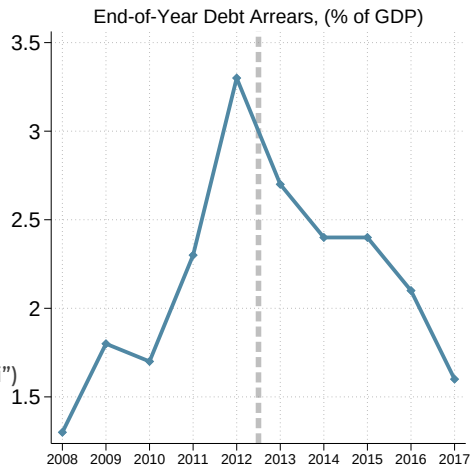
- ◇ Commercial debt=5.8% of GDP, $\approx 57\%$ overdue

- ▶ 2011: European Union directive

- ◇ Requirement to settle invoices in 1 month

- ▶ **2013 decree by Italian government** ("Sblocca Debiti")

- ◇ €57bn to public administrations to clear backlog



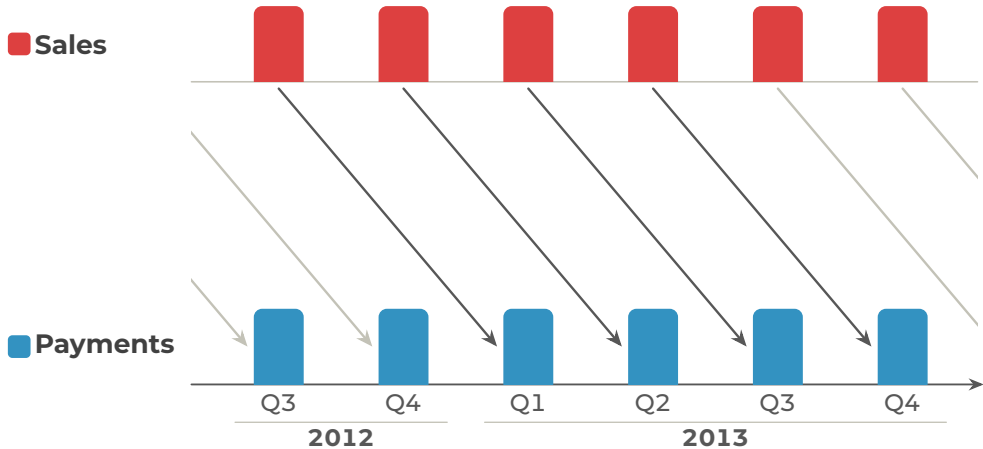
WHY THIS GENERATES A CASH-FLOW SHOCK

► Institutional design rules out a demand shock

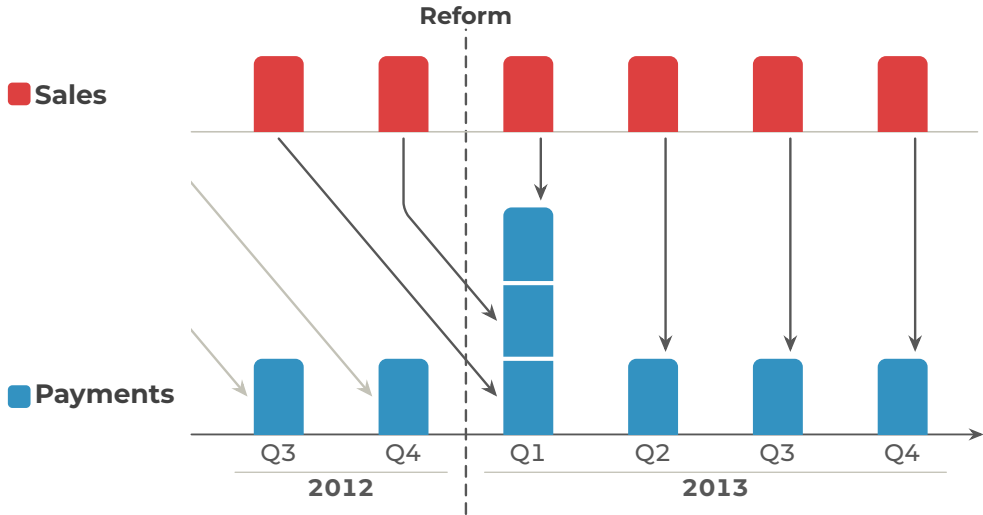
- ◇ Funds earmarked for **pre-existing** claims (31 Dec 2012)
- ◇ Funds must **immediately** extinguish listed debts,
- ◇ **Loans from central government, not transfers**, no additional spending capacity
- ◇ **No response of procurement spending** for public bodies with more allocated resources

→ shock to firm liquidity, not to firm demand

WHY THE REFORM GENERATES A CASH-FLOW SHOCK



WHY THE REFORM GENERATES A CASH-FLOW SHOCK



EMPIRICAL ANALYSIS

WHAT WE NEED

► We need:

- I Firm-level data on investment
- II Measure of exposure to the reform
- + Characterize MPI heterogeneity → large sample

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► We need:

- I Firm-level data on investment
 - II Measure of exposure to the reform
 - + Characterize MPI heterogeneity → large sample
- leverage admin data on the universe of **firms** and **public procurements**

DATA

- ▶ **Balance sheet + income statements** data on universe of Italian LLCs (chambers of commerce)
 - ◇ Outcomes: investment, liquidity, trade receivables, ...
 - ◇ Also: province, leverage, 2-digit NACE sector...
- ▶ Quasi-universe of **public procurement contracts** from Italian PAs (anti-corruption authority)
 - ◇ Identifiers for awarding body and winning firm ($\approx 30k$ firms) + contract value
 - ◇ 6.9% of GDP, 36% of government consumption
- ▶ Scraped data on **reform allocations per public body** (Italian Ministry of Finance)
 - ◇ Observe each body's ($\approx 20k$) exposure to the policy

CONSTRUCTING THE SHIFT-SHARE INSTRUMENT

- **Shift:** intensity of treatment of body k :

$$g_k = \frac{\text{reform resources to settle debts of body } k}{\text{pre-2013 procurement value of body } k}$$

- **Share:** firm i 's pre-2013 reliance on body k :

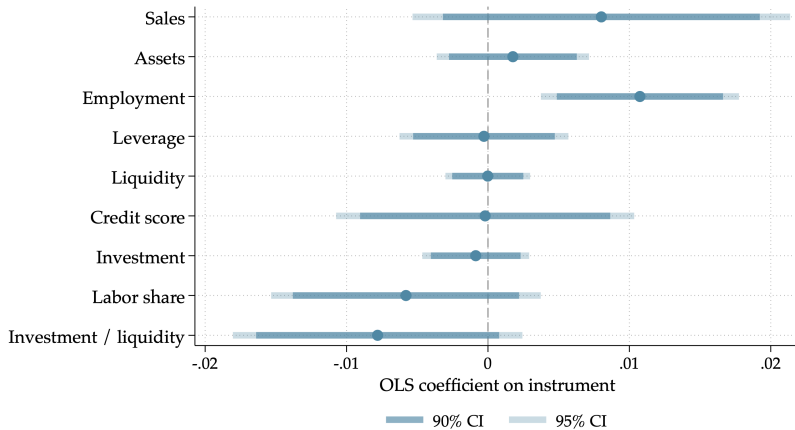
$$s_{ik} = \frac{\text{Sales}_{ik}}{\sum_{k'} \text{Sales}_{ik'}}$$

- **Instrument:** exposure-weighted average shift:

$$z_i = \sum_k s_{ik} g_k$$

BALANCE TEST

$$y_{is} = \alpha + \beta z_i + u_i, \quad s = 2012$$



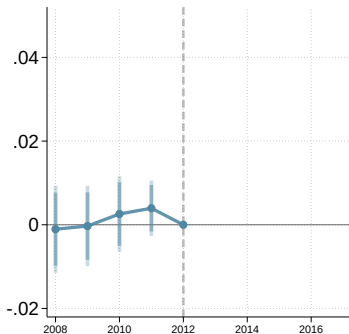
FIRST STAGE: A LIQUIDITY SHOCK

$$\log y_{it} = \alpha_i + \delta_t + \sum_{s \neq 2012} \beta_s (\mathbf{1}_{s=t} \times z_i) + \varepsilon_{it}$$

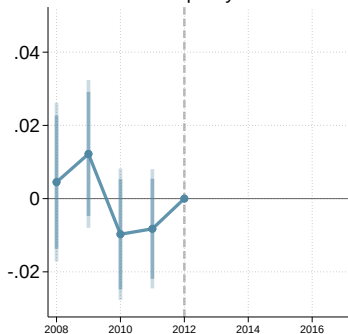
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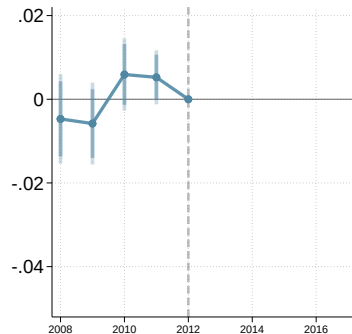
Sales



Liquidity



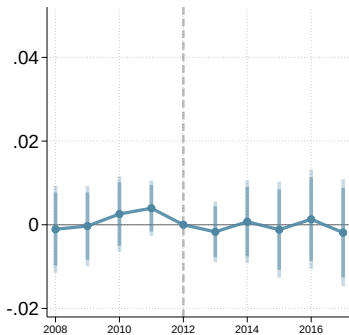
Trade receivables



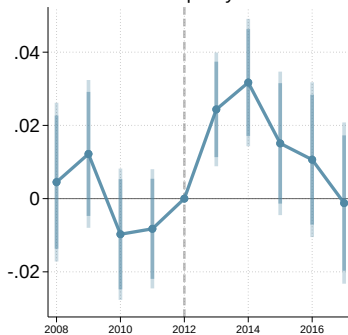
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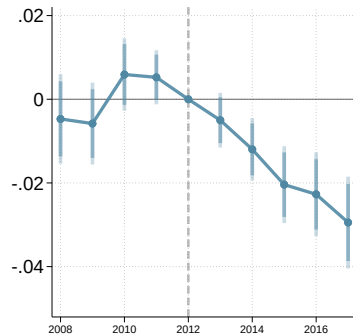
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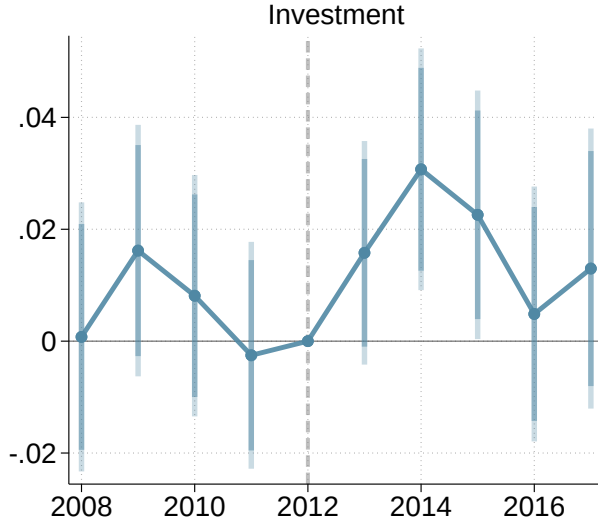
Liquidity



Trade receivables



REDUCED FORM: INVESTMENT RISES

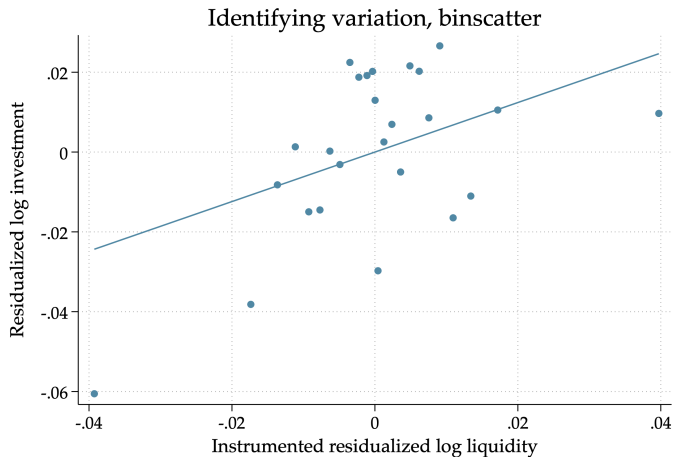


IV ESTIMATES OF THE MPI

$$\log I_{it} = \alpha_i + \delta_t + \gamma \log \widehat{\text{Liquidity}}_{it} + u_{it},$$

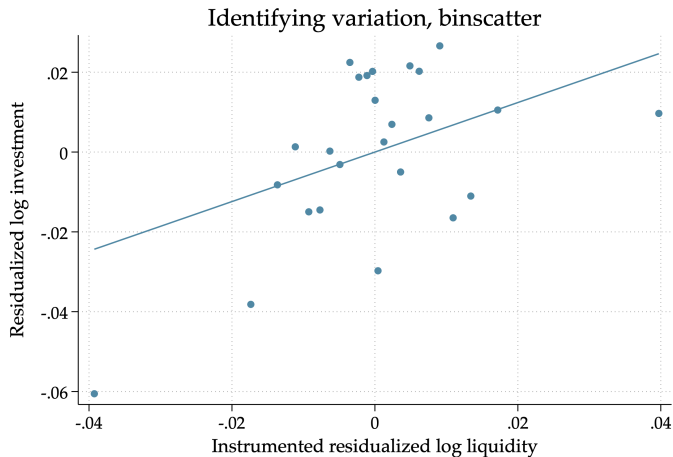
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$$\log I_{it} = \alpha_i + \delta_t + \gamma \log \widehat{\text{Liquidity}}_{it} + u_{it}, \quad \widehat{\text{MPI}} = \hat{\gamma} \times \varrho, \quad \varrho \equiv \mathbb{E}I_i / \mathbb{E}L_i$$



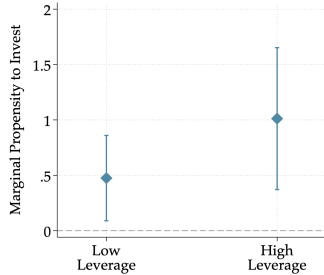
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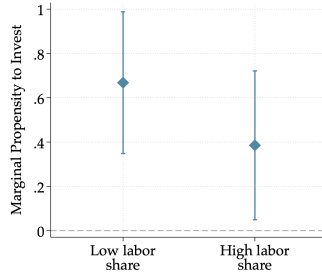
	Narrow Window (2011–14)			Broad Window (2010–15)		
$\log \text{Liquidity}_{it}$	0.775*** (0.233)	0.758*** (0.243)	0.680** (0.299)	0.767*** (0.195)	0.737*** (0.215)	0.620** (0.260)
KP F-stat.	25.09	22.67	14.26	31.99	25.81	15.89
Observations	93,452	93,416	93,409	141,466	141,414	141,404
Year FE	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓
Province×Year FE		✓	✓		✓	✓
Sector×Year FE			✓			✓
MPI	0.526	0.514	0.527	0.548	0.527	0.503

HETEROGENEITY

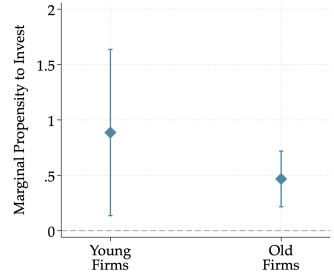
Along leverage



Along labor share



Along age

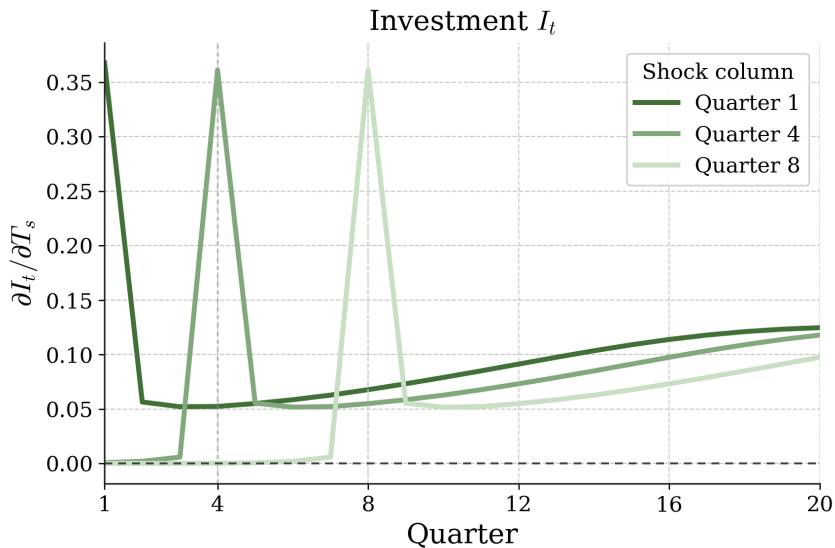


MODEL MEETS DATA

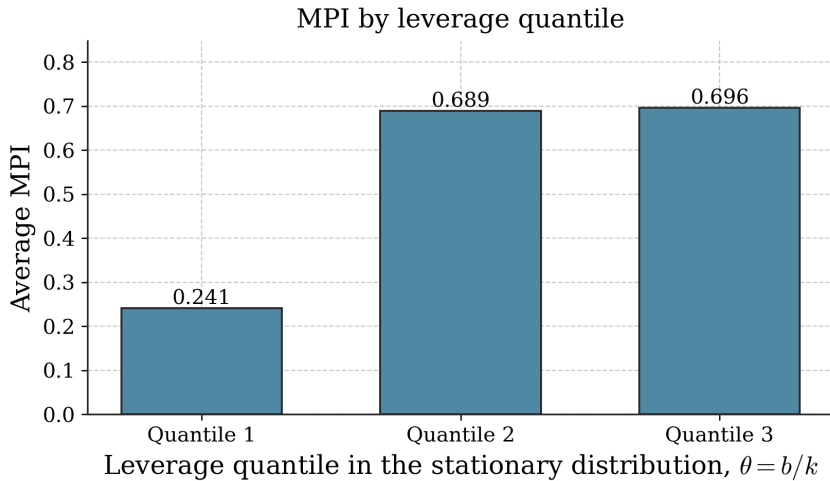
CALIBRATION

Parameter	Description	Value	Target / role
<i>Production and depreciation</i>			
α	Capital exponent	0.32	Returns to scale
ν	Labor exponent	0.60	Labor share
δ	Quarterly depreciation rate	10% annual	Investment-capital ratio
<i>Discounting and adjustment costs</i>			
R	Interest rate	2% annual	Standard
ϕ	Quadratic adjustment cost	1.5	Chodorow-Reich (2026)
<i>Financial constraint and exit rate</i>			
χ	Collateral constraint, $b' \leq \chi k'$	0.38	Match MPI
T_f	Steady-state firm transfer	0	Baseline normalization
θ_{exit}	Quarterly exit probability	0.02	Firm exit rate
<i>Idiosyncratic productivity</i>			
ρ_z	Persistence of $\log z$	0.94	Syverson (2011)
σ_z	Innovation s.d. of $\log z$	0.085	Syverson (2011)

MODEL MEETS DATA



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Takeaways

- ▶ Causal estimates of MPI: firms invest €0.50 of extra €1 cash flow over a year
- ▶ Concentrated in **leveraged** firms, **capital-intensive** sectors and **younger** firms

Open questions

- ▶ Quantitative model: from the MPI cross-section to aggregate & GE effects
- ▶ Optimal design of liquidity-based stabilization

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